

Exit Strategy for UK Leveraged Buyouts: Empirical Evidence on Determinants

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Abstract

This study provides evidence on the determinants of exit strategies for UK leveraged buyouts (LBOs). Using a sample of 104 going concern UK LBOs exited by the LBO sponsors between January 1998 and June 2004, we analyse the determinants of the exit routes – IPO, trade sale or secondary LBO - by estimating a trinomial logistic model. Operating performance, firm size, length of holding period by the LBO sponsor and whether the firm belongs to the ‘high-tech’ industry are significant determinants of the exit strategy. Operating performance is significantly higher in the secondary buyout sample than in the other two categories, and holding period is also significantly higher in secondary buyout than in trade sale samples. Though the evidence is not quite conclusive, it suggests that when exit markets are not favourable, LBO sponsors hold on to better performing firms longer, in anticipation for favourable exit market conditions, while disposing the others through trade sale and flotation. This is the first study indicating that the secondary LBO market might be ‘sieving-off’ the best companies from the portfolios of LBO sponsors. I also find evidence, contrary to existing theory, that exit returns have no significant predictive power for the exit route, and that larger firms are less likely to be exited by flotation and trade sale. These preferences of LBO sponsors are likely to be the effect of unfavourable exit market conditions.

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1. Introduction

A leveraged buy-out (LBO) is an acquisition of a corporation as a whole or a division of a corporation mostly with cash, the cash being raised with a preponderance of debt raised by the acquirer. A special purpose company, called Newco is created to carry out the acquisition. This acquisition vehicle's financial structure consists of substantial debt, sliced into different types of debt, and equity. The equity component is provided by investors who invest in a 'fund' raised by a specialist firm that finds suitable investment opportunities to invest the fund through one or more Newcos. Since the equity capital is contributed by investors privately, outside a public stock market, this source of finance is referred to as private equity and the specialist firm is called a private equity firm. The leveraged buyouts could be any of the following; distressed or out-of-favour public companies, divestitures that result from the sell-off of the divisions of a public corporation, or privately held and family-controlled companies in which the owners are willing to relinquish interest.

LBOs became prominent in the 1980s when the industry experienced remarkable boom, particularly in the United States where there were over 1,400 'going private' transactions between 1981 and 1989. In the United Kingdom, LBOs only began to grow rapidly in the 1990s, with about 3,838 deals between 1990 and 2002, out of which only 152 were 'going private' deals (Sudarsanam, 2003, ch.12).

There are several factors that are considered critical success factors for LBO specialist firms (also known as LBO sponsors), but most worthy of note are fund raising ability (mostly determined by credibility of the sponsor) and capacity to add value (a key deal-winning factor), and ability to achieve successful exits and meet the obligation to return funds to their investors at the right time, while delivering a respectable return on investment. Private equity firms seem to have been reasonably successful in fundraising, with consistently high levels of fundraising activity in Europe and the UK in particular, over the few years, at times resulting in too much money chasing too few

deals. They have however faced tremendous challenges with regards to exiting their investments for almost a decade now.

Following the private equity boom days of the 1980s are the 'black days' since the mid to late 1990s characterised by exit difficulties resulting from illiquid capital markets and global lull in the mergers and acquisitions market. Investment holding period has been on the increase as a result of this exit problem, thereby causing concern to many private equity operators over the past decade, particularly in Europe. The European Venture Capital Association (EVCA), way back in 1996 commissioned PricewaterhouseCoopers (PwC) Corporate Finance team to conduct a survey aimed at improving the investment overhang¹ situation, by educating the venture capitalists on planning and executing their exit strategies. The success of an LBO investment, and in fact venture capital in its entirety, is dependent on successful exit because that is when the value created by the LBO is realised. LBO firms' ability to exit in good time ensures they meet their repayment obligations (principal and returns) on maturing funds. Smith and Wall (1998) commenting on the investment overhang problem conclude that venture capitalists appear to be better at investing than they are at exiting.

Now, turning specifically to the UK, the LBO market experienced significant decline in value in recent times. The Centre for Management Buy-out Research (CMBOR) which estimates the UK LBO market at about £16.2billion as of 2003, down from £23.9billion in 2000, has consistently reported the difficulty of LBO exits since the year 2000, with this trend persisting to date. As shown in Figure 1 below, trade sale has always been the most common exit route in the UK even before the global capital market decline of 2000. While exits through flotation were sizeable, ranging between 35 and 48 from 1993 to 1997, they have dropped to below 20 since 1998, and conspicuously constitute a much lower percentage of all London Stock Exchange flotation as shown in Figure 2. The persistence of this problem, poses a potential threat to the private equity industry and the LBO firms in particular, as it mounts significant pressure on their return on investments.

The growing reputation of their business as an illiquid one could lower their default risk rating, attracting remarkably higher risk premium demand from their

¹ significant number of investee companies awaiting to exit

financiers, while the pressure to exit investments at the end of the financing cycle may result in forced sales at significant discounts. Should this trend persist, the investment and entrepreneurial community stand to lose the immense benefits they currently derive from the expert advisory and corporate governance roles the private equity firms currently play alongside their financial arbitrage activities. Even the economy at large would be negatively impacted by a significant decrease in private equity activities, as their financing role fosters economic development. This study therefore seeks to contribute to the currently limited understanding of LBO exit strategy by providing empirical evidence to inform exit strategy formulation, and thereby facilitate better exits.

So far, most of the studies on LBO exit strategies and returns are based on US data; with few on European data. While The CMBOR provides valuable comprehensive statistics of UK LBO deals dating far back to the 1980s, there are hardly any studies in this area using UK data. The UK LBO market at £16 billion is of considerable size, and therefore a better understanding of the relative strength of factors driving the exit route would be vital for private equity practitioners in analysing potential investments, and would also help them prepare their portfolio companies for exit. Further more, since LBO markets differ from country to country on a number of dimensions, including exit route as highlighted by Sudarsanam (2003), the results of this study should also be of much benefit to foreign private equity firms seeking to make an in-road into the UK LBO market.

Figures 1 and 2 here

We examine a sample of 104 LBO firms exited by their sponsors through one of the three exit routes – stock market flotation through an initial public offering (IPO), trade sale to a strategic corporate buyer and sale to another LBO sponsor, called secondary buyout (SBO). The exits occur during 1998-2004. We model the exit route choice as a function of the characteristics of the LBO firm, the sponsor, the capital markets and the mergers and acquisitions (M&A) market. Our results show that firm size, its operating performance in the run-up to the exit, leverage, how long the investment is held by the sponsor between the original LBO and exit, and the high-tech industry affiliation of the firms are factors that significantly influence the exit route choice. We find no significant difference in the exit returns of the different routes. There

is some weak evidence that high tech LBO firms are more likely to exit by an IPO. Firms with the strongest operating performance and held longest by the sponsor are more likely to be sold to another LBO sponsor. Thus the exit decision is systematically related to the firm attributes and investment policy of the sponsor. We highlight the implications of these results for LBO firms that need to evolve optimal exit strategies.

The remaining part of this paper is organised as follows. Section 2 develops the conceptual framework and formulates the research questions. Section 3 describes the methodology and data. Results are presented in Section 4 and the final section provides a summary and conclusions.

2. Review of prior literature

2.1 Role of LBO specialist firms and sources of value creation

Ever since LBO specialist firms became prominent in the early 1980s, they have augmented stock market efficiency by exploiting arbitrage opportunities in firms undervalued by the market - buying them at the undervalued rate, and subsequently disposing them at a much higher valuation realising super returns in the process. Such arbitrage is widely believed to be responsible for their phenomenal growth in the 1980s. But as the private equity market has evolved and become highly competitive, LBO sponsors² have widened their strategies to become key *agents of corporate turnaround and reorganisation*. Today, the business logic of most LBO transactions lies in value creation through turnaround, operational efficiency and optimal utilisation of corporate resources, within the framework of an enabling capital structure. There is very rich literature³ on the sources of value creation by LBOs, and most of them revolve around the core turnaround strategies extensively discussed in Sudarsanam and Lai (2001).

Sources of value creation in LBOs

The various sources of value from LBOs include:

² LBO sponsors are private equity firms specialising in leverage buyout transactions.

³ Berg et al (2003) provide a comprehensive literature review on value creation by LBOs.

- tax benefits from high leverage (Kaplan, 1989a; Travlos and Cornett, 1993);
- reduced agency cost through close monitoring by the private equity investor and lenders and through performance-related managerial incentives e.g. stock options and stock grants (Jensen, 1986);
- Efficiencies from asset and operational restructuring (Smith, 1990; Kaplan, 1989b; Holthausen and Larcker, 1996);
- Effective corporate governance structures through LBO sponsor's board representation (Lerner, 1995).

We now turn to how value created by LBOs is measured. As we argue, value created or return to sponsors is not independent of the exit route for the sponsor to unwind its investment.

Measuring value created by the LBO

The value of an LBO firm at the point of exit of the sponsor compared to its valuation at the time of the buy-out should give a fair indication of the value created by the LBO. Therefore the exit return to the LBO sponsor should be a reasonable proxy of the value the LBO has created. Various prior studies have compared private equity returns to stock market return from public companies, while a few others have reported private equity returns from different exit strategies. These studies explain their results as a consequence of a liquidity risk premium and under-pricing due to information asymmetry.

Ljungqvist and Richardson (2003) using a unique US dataset of private equity funds show that private equity generates excess returns on the order of 5 – 8% per annum relative to the aggregate public equity market. They argue that this represents compensation for holding a 10-year illiquid investment. Kaplan and Schoar (2003), however, find that average fund returns are about the same as the S&P 500. While some studies have reported substantial excess returns from private equity investments above the public market index returns, it is often argued that the 'excess return' is actually a risk premium reflective of the higher level of risk borne by private equity investors. According to Das et al (2003), investing in private equity is akin to buying a highly risky discount security, where the maturity date is unknown, therefore with substantial pay-off risk.

On the other hand, Emery (2003), using US data from Venture Expert, finds higher risk adjusted private equity returns from un-exited investments (corresponding to higher Sharpe ratios⁴) than public equity returns, but also records a higher correlation between private and public equity returns after adjusting for 'stale pricing'⁵ by using longer horizon (annual and biannual) data as against the quarterly returns data previously used⁶. Therefore results from studies using such data should be scrutinised for the stale pricing problem. Weideig and Mathonet (2004) analyze the risk profiles of direct investments in portfolio companies and investments in private equity funds from 1980 to 1998 in detail. They conclude that there is a diversification benefit for funds and funds-of-funds, and that the risk profile of a fund is relatively symmetrically distributed with a 30% probability of total loss of capital invested.

A very recent European study on returns is that of Kaserer and Diller (2004). They present a cash flow-based analysis of the return and risk characteristics of European Private Equity Funds. They document a cash flow based IRR of 12.7% and an excess-IRR of 4.5% relative to the MSCI¹ Europe equity index over the period 1980-2003. They also note that over the same period, private equity funds generated an over performance with respect to both the MSCI¹ Europe equity and the J.P Morgan Government Bond indexes.

2.2 Determinants of LBO exit strategy

LBO sponsors realise their returns by exiting their investments, whether in response to market volatility, regulatory changes, corporate restructuring, or fiduciary obligations to their own investors. Typical exit routes include trade sale to a strategic buyer, initial public offering (IPO) and sale to another financial buyer in a secondary LBO. Other possible exit routes are; re-capitalization, which allows equity holders to realize a return by taking a sizable dividend and through receivership/liquidation where the buy-out firm fails.

⁴ Sharpe ratio is a measure of risk adjusted excess returns developed by William F. Sharpe. It shows the reward per unit of risk, and is given as the periodic return, less the risk free rate, divided by the standard deviation of the periodic returns.

⁵ The stale pricing problem is the delay in adjusting Net Asset Values of portfolio companies (un-exited investments) to market, by private equity general partners.

⁶ Stale pricing problem results in either an understatement or overstatement of net asset values, giving a false impression of return on assets. This is worthy of note particularly with the self-reported private equity quarterly returns held on Venture Expert.

The funds invested in LBOs have a pre-determined life, usually about 5 to 10 years, and the LBO sponsors have to repay the limited partners and recycle their own investment capital to sponsor future LBOs (Sudarsanam, 2003, ch.12). Therefore, the assessment of possible exit options and the articulation of a clear and feasible exit strategy are of paramount importance to LBO sponsors during the pre-investment analysis stage. The success of an LBO for a private equity firm is generally measured by the ease of exit and the amount of realised returns on their investment. Based on the foregoing, the quest for a sound understanding of the factors that drive the choice of LBO exit route has continued to motivate various academics and industry practitioners to research in this subject area.

Detailed review of the literature shows that the choice of exit route from an LBO could be influenced by the interplay of various factors. Various studies, mostly using US data have documented most of the factors. While most of the studies are not specifically based on LBO samples, the underlying drivers of the exit decision are equally applicable to LBOs. In the following subsections we discuss these factors.

Liquidity of exit markets

Private equity firms seem to follow a pecking order in their choice of exit route from their investments; first through an IPO, then a trade sale if an IPO is not possible and then other less preferred routes like the secondary buyout. Thus in the private equity literature, the concept of liquidity is generally used to mean the possibility of exiting either by flotation on a stock market or a trade sale in the corporate M&A market. Therefore the state of the capital market, investor sentiments and the acquisitive appetite of trade or strategic buyers, which are largely conditioned by the state of the economy, are crucial to providing liquidity for LBO exits. Black and Gilson (1998) emphasise the importance of a well developed capital market to the existence of a vibrant venture capital market.

Lerner (1994) using a sample of 350 privately held biotechnology firms financed by venture capitalists between 1978 and 1992 shows that venture capitalists successfully time IPOs and are more likely to take companies public when their valuations are high under favourable market conditions. Brau et al (2003), show that the relative “hotness” of the IPO market enhances the probability of an IPO. Cumming et al (2003) argue that

when markets become less liquid, venture capitalists (VCs) tend to invest more in new early-stage projects in order to postpone exit requirements and, in so doing, invest in riskier projects. Golbe and White (1993) in their study which provides evidence that U.S mergers come in waves, argue that favourable corporate M&A market conditions can encourage bidder activity. Therefore, activity on the corporate M&A market could increase the propensity to exit through trade sales.

In the UK, the exit market liquidity situation has remained a source of concern. CMBOR (2004a) reports the continued growth of the secondary buy-out market over the past 4 years and argues that it serves as the main source of liquidity for the private equity market in the UK in the absence of suitable capital market condition.

Operating performance

Operating performance is certainly a crucial factor in corporate valuation, as it often forms the base for cash flow forecasts, and testifies to the strength of the business fundamentals. Therefore, intuitively it should be vital to the consideration of the exit route, particularly for an IPO. Except when there is strong persuasive evidence to support expectation of better future operating performance and hence higher cash flow generation, offering a company with poor recent or current operating performance for sale would result in lower valuation and consequently lower returns. Degeorge and Zeckhauser (1993) find strong evidence that return to public ownership of reverse LBOs coincides with a peak in their operating performance in the pre-offering year. Bienz (2004) argues that highly profitable companies that require little oversight go public whereas less profitable companies that require more control are sold off via trade sale.

Exit returns

Since the ultimate objective of the LBO sponsor is profitability (return on investment), and this is only realisable on exit, it would be expected that returns realised from different exit routes would be a critical factor considered in the decision to exit. Smith and Wall (1998) from their survey of VCs in Europe, report that “most VCs view price, or IRR, as the over-riding factor determining their approach to exit. Where the VCs mentioned other factors, it was often because of their impact on the IRR”. They also find that the VCs generally preferred exit through an IPO as it tended to yield the

highest returns. Bienz (2004) provides empirical evidence that IPOs have higher rates of return than trade sales. He uses a sample of 423 trade sales and 108 IPOs obtained from the CEPRES⁷ database in Frankfurt, and covering the period from 1971 – 2003, finds a significantly higher IRR of 123.42% for IPOs as against 75.32% for trade sales.

Reputation of private equity firm

Irrespective of the due diligence and the comfort provided by reporting accountants during an IPO, investors find assessing privately owned companies in a flotation challenging due to information asymmetry⁸. There is considerable literature and evidence to support the theory of under-performance of venture-backed IPOs due to information asymmetry, which the market penalises through significant discounts. Hogan et al (2001) report a mean excess return of 7.64% during 1987 – 1998 for reverse LBOs as against 13% for original IPOs, supporting the information asymmetry hypothesis.

Disposing of companies in the course of investment exits is a repeat game for private equity firms particularly in the IPO and the secondary buyout markets (where they are likely to deal with fellow private equity investors). Therefore building up reputation is essential, particularly as it provides additional comfort to investors regarding the strength of business fundamentals and valuation of the companies brought to the market. Various studies have shown that the reputation of the private equity firm helps to reduce ex-ante uncertainty and mitigate the costs associated with information asymmetry in the IPO market. Thus high-reputation LBO sponsors ‘certify’ the quality of the IPOs they bring to the market.

Neus and Walz (2004) describe the reputation of private equity firms as the trust outside investors place in the pricing of a certain issue, and articulate two important aspects of reputation in the context of IPO exits. “On the one hand, VCs, especially young ones, have a strong incentive to build up a reputation with their investors by means of realising a successful IPO with high returns as soon as possible. This facilitates refinancing the VC (especially for first time funds) when it comes to raising

⁷ Center of Private Equity Research

⁸ Condition in which at least some relevant information is known to some but not all parties involved.

new money for a follow-up fund. On the other hand, VCs may have a strong incentive, when playing the IPO market more than once, to establish a reputation for being an honest partner, i.e. not disclosing any false information or, equivalently, not mis-pricing the issue.” Gompers (1996) demonstrates that the building of reputation particularly for younger venture capital firms motivates the decision to go public. Black and Gilson (1998) and Sahlman (1990) provide further support to this motive. But in a recent UK study, Wright et al (2004) found that MBOs backed by highly reputable VCs tend to be older companies, exit earlier and are more underpriced than MBOs backed by less reputable VCs. Their results contradict the ‘certification’ hypothesis supported by US data (Megginson and Weiss, 1991 and Gompers, 1996).

Industry prospects

Investors value highly firms in certain industries at certain times e.g. frenzied investments, during the internet bubble period in the late 1990s, in the IPO of “dot.com” and high tech companies. According to Allen and Gale (1999), given the popularity of IPOs among investors over this bubble period, privately held high-tech firms were more likely to capitalise on this enthusiasm by choosing an IPO instead of pursuing any other exit route. Brau et al (2003) find that high-tech industry affiliation is positively related to the probability of an IPO. Mitchell and Mulherin (1996) find significant differences in the rate and timing of merger activity across industries. Akhigbe et al (2003) document various industry effects that influence the timing and pricing of an IPO.

Highly fragmented industries where there is scope for significant economies of scale benefits like financial services, retailing and leisure are more likely to be targets of strategic acquirers seeking to achieve consolidation than exit via an IPO. This hypothesis is supported by Berger et al (1995) and Berger et al (1999). According to CMBOR (2004), the service sectors dominate the buyout market in the UK, with leisure, retail and business services sectors at £9.8 billion, accounting for over 60% of market value in 2003. Since these industry sectors are traditionally highly fragmented, they are likely to be more prone to consolidation and a trade sale to a strategic buyer may be preferred as an exit route.

Investment holding period

The length of time an investment has been held in the portfolio of a LBO sponsor could have an influence on the exit route. The average life span of a UK buyout has gradually increased since the beginning of the 1990s, as indicated by the average time to exit increasing to just over five years in 2001, as compared to 3.5 years at the start of the 1990s (CMBOR, 2004b). This is very much the same situation in the rest of Europe, where assets held by buyout funds for more than four years rose from 42% in 2000 to 66% in 2003 (Weir, 2004). Sudarsanam (2003, ch.12) advances an explanation for increasing trend in holding period. He maintains that buyout firms need time to achieve operating performance and cash flow improvements rather than rely on restructuring, downsizing and divestitures to raise non-operating cash flows and pay down debt. Such operating performance improvement is a long haul thereby lengthening the holding period.

In a pioneering study of the relationship of the holding period with industry and exit route, Giot and Schwienbacher (2004) modelled the exit times of about 6000 venture-backed firms using completing risk models. Their study covers three exit channels; IPO, trade sale and liquidation. They report that as time flows, venture capital-backed firms first exhibit an increased likelihood of exiting by an IPO. However after having reached a plateau, investments that have not yet exited have fewer and fewer possibilities of IPO exit as time increases. They also find that the industry type matters to the timing of exit, as biotech firms have the fastest IPO exits, and internet firms are the fastest to liquidate, while biotech firms are the slowest.

Since LBO sponsors are financial investors with investment holding periods constrained by the time to expiry of their funds, they are likely to exit from their investments at the earliest possible time that maximises their returns. The longer the holding period may be an indication of the difficulty of more attractive and higher value exit routes. To the extent that many venture capitalists consider the IPO route as the most attractive exit route, a longer holding period suggests this option was not available and that the LBO sponsor had to follow other routes.

Firm size

Pagano et al (1998) find that the size of a company is significantly correlated with the probability of listing, arguing that larger companies are more likely to go public. The size of a firm can also provide an indication of its ability to survive the cost of going and staying public. Pagano and Roell (1998) argue that IPOs involve high fixed costs, which make an IPO a relatively costly exit option for small firms, consistent with the survey evidence provided by Smith and Wall (1998). Furthermore, whereas small firms may not be equipped to successfully operate as stand-alone public companies, they can provide value for acquiring firms who could share their own resources and skills with the target firm (Brau et al, 2003). Smaller companies are unable to bear the cost of flotation, withstand the onerous regulatory requirements expected of public companies, and might not be able to compete favourably as quoted companies. They may therefore have a higher propensity to be exited through either a trade sale or a secondary buyout.

Financial leverage

Firms with high debt and related interest costs may consider an IPO as a means of changing their capital structure and subsequently gaining access to cheaper funding. This theory finds support in the study by Pagano et al (1998) in which they document a reduction in the cost of bank credit for independent companies after an IPO. Equity investors may also feel that for high leverage firms the problem of adverse selection is mitigated by the close monitoring by lenders. Therefore high leverage firms may be attractive IPO candidates (Brau et al, 2003). Firms with high debt and related interest costs may also consider flotation as a means of changing their capital structure and subsequently gaining access to cheaper equity funds. On the other hand, investors may perceive high debt level as a potential risk factor and therefore apply significant discounts in the pricing of such companies. Going by this argument, higher leveraged firms may opt for more conservative restructuring path, i.e. a takeover (trade sale).

Motivation for going private

Public companies have various reasons for going private. For instance, firms whose managers believe they are undervalued by the stock market may decide to go private to obtain a better valuation for the firm's shares. Also, management may be tired of keeping up with the short-termism of public shareholders or unwilling to bear the regulatory burden of disclosures, strict corporate governance rules etc. The introduction of Sarbanes-Oxley Act for instance has imposed new costs and obligations on publicly quoted companies. Many small companies have complained of the excessive regulatory costs of public listing.

The UK capital market has seen an increasing number of companies going private in the past decade with a continued net reduction in the number of companies quoted on the London Stock Exchange to about 1600 in 2003 compared to over 2000 in 1998 (CMBOR, 2004a). Conceivably, the decision on the exit route from a going private LBO would be influenced by the motivation of the firm for going private in the first place. For such companies, flotation (reverse LBO) is not a likely exit route, since they would not want to continue facing the 'irritable' conditions that necessitated their going private in the first place.

In public-to-private LBOs occasioned by one or more of these factors, management may be opposed to exit through an IPO. Smith and Wall (1998) note from their survey of private equity practitioners, that management sometimes resist a trade sale for fear of losing their jobs. In such instances, a secondary buyout may be management's preferred option. But Kaplan and Stromberg (2003) show that venture capitalists include control rights and covenants in their contracts to keep their exit options open.

Smith and Wall (1998) summarizes the advantages and disadvantages of different exit routes from the practitioners' point of view, based on a survey conducted for the EVCA on 30 venture capitalist firms across Europe⁹. Their report shows transaction costs and ease of deal completion as other important factors considered by LBO sponsors in the choice of exit route.

⁹ See appendix 1.

There is no prior UK study that has examined such a comprehensive set of factors in determining the ultimate exit strategy of LBO sponsors and neither has there been any, evaluating the relative exit returns from these three exit routes. This study seeks to fill this void. The UK is the most active private equity market after the US but also has institutional differences with the latter. This study therefore presents an opportunity to test whether the exit strategies pursued by UK firms differ from those of the US firms.

Overall, the exit route decision of the LBO sponsor is determined, *ex ante*, by a variety of factors including the antecedents of the target firm, its characteristics, attributes of the sponsor and external market conditions. With some of these factors such as small firm size, prior existence as a public company, high liquidity of the IPO market and high historic returns from IPOs we expect that the probability of an IPO exit are higher than that of the other two exit routes. In the case of other factors such as operating performance, high leverage, sponsor reputation, industry affiliation and the holding period there is no clear cut prediction of the exit route. In these cases prior empirical literature, albeit from non-UK markets offers conflicting evidence. We look to our empirical analysis to resolve some of these conflicts. We model the exit route decision as a function of these factors.

3. Methodology and data

3.1 The logistic model of the exit route

The exit probability models are based on a trinomial logistic regression using the three identified exit routes as the response (dependent) variables.

$$\begin{aligned} \text{EXITROUTE} = & \alpha + \beta_1 \text{MKTLIQ} + \beta_2 \text{HTINDUST} + \beta_3 \text{FSIZE} + \beta_4 \text{PEREPUT} + \beta_5 \text{PTPS} \\ & + \beta_6 \text{FINLEVER} + \beta_7 \text{OPPERF} + \beta_8 \text{HPERIOD} + \beta_9 \text{EXITRTN} \\ & + \beta_{10} \text{CONINDUST} + \varepsilon \end{aligned}$$

where EXITROUTE represents exit route and is denoted as 0 if IPO, 1 if trade sale, and 2 if secondary buyout. The explanatory variables are defined as follows:

MKTLIQ = exit market liquidity;

FINDUST = firm industry;

FSIZE = firm size;

PEREPUT = reputation of private equity firms;

EXITRTN = exit return;

FINLEVER = firm's financial leverage

HPERIOD = the period during which the investment is held by the LBO sponsor;

OPPERF = operating performance;

PTPS = 1 if the firm was earlier a public-to-private LBO and 0 otherwise and

ε = the error term.

Exit return (EXITRTN)

As proxy for historic exit returns derived from particular exit routes, we use the exit returns realised by the LBO sponsors on the sample deals. Though they are *ex post* estimates and would not have been known at the time of the decision regarding the exit route, we have used them because they are the only available indication of historic returns. Moreover, the financially sophisticated LBO sponsors and their financial advisers are in a position to make reasonably accurate estimate of expected exit returns through different routes.

Exit returns are measured as annualised exit multiples as in Das et al, (2003). As pointed out by Weideig and Mathonet (2004), exit multiple is the simplest and most intuitive way to measure exit returns, as it expresses how much money was received back as a multiple of the amount paid in.

$$\text{Exit multiple } X_i = (\text{Exit valuation} / \text{Entry valuation})$$

Because various LBOs are held for varying periods, a comparison would only be fair if the returns are standardised. We therefore normalise the multiple for differences in holding period (between entry and exit) by annualising the multiples using the formula:

$$X_{\text{annual}} = [X_i]^{1/t} \quad \text{where } t = \text{day}/365$$

The annualised return is then obtained by the formula, $R_{\text{annual}} = (X_{\text{annual}} - 1)$, as in Das et al, (2003).

3.2 Data for other explanatory variables

We use the total asset value of the firms reported in the most recent annual financial statement just before the exit date as proxy for size. The first two digits of the UK SIC 2003 primary industry classification codes are used to classify the sample firms into different industries. These two digit codes represent broad groups of related business activities. Where the acquiring company is not a UK company, we obtained details of their primary activities from the company's website and used this as a basis to allocate an appropriate UK SIC code. In view of the industry classification effects discussed in the literature review (Braun et al 2003, CMBOR 2004), we also employed an alternative categorisation of the sample into high-tech, and consolidation-prone industries (financial Services, retail and leisure).

For stock market liquidity, we use the London Stock Exchange (LSE) IPO volumes, i.e. the number of IPOs on the LSE in each of the affected years as proxy for exit market liquidity as in Cumming et al (2003). In their study of liquidity of exit markets, they used the number of IPOs per year on the NASDAQ, NYSE and AMEX as proxy for liquidity of exits. Jenkinson and Ljungqvist (2001) denote this same proxy as the annual "IPO volume." However, we also use the pound sterling value of IPO funds as an alternative proxy to capture the extent of investment appetite of the investment community in the stock market in our robustness tests. While Cumming et al (2003) argue that IPO volumes are also a good measure of corporate M&A activity, we use the deal flow (volumes and values) in the UK corporate M&A market, extracted from Thomson's SDC Platinum as proxy for liquidity of this market where LBO trade sale exits occur.

We use the earning before interest, tax, depreciation and amortisation (EBITDA) divided by total asset as the measure of operating performance, as in Zeckhauser and DeGeorge (1993) and Kaplan (1989a). This is robust to noises introduced by accruals and discretionary accounting policies for depreciation and amortisation; particularly in an LBO where fixed assets and good will are typically re-valued and their book values written up to reflect the price paid for the firm's assets. This inevitably leads to subsequent increases in reported depreciation and amortisation expense, reducing reported income after the buyout. Total debt to total asset is used as a measure for financial leverage. Total debt includes short and long-term debt. This is robust to the

problem associated with using debt to equity ratio, as a number of the companies in the sample had negative shareholders' fund. Our criterion for reputation of the LBO sponsors is the worldwide ranking of private equity firms by size (defined as the number of investment professionals), produced by Pseps Limited, publishers of 'The Private Equity and Venture Capital Directory'. The ranking is publicly available (online) at their website (www.private-equity.org.uk). This criterion assumes that the number of investment professionals a firm has could be proxy for the volume of its business, which is as a result of its goodwill and reputation in the market place. Though this logic may not always hold, it seems the most objective and independent ranking available.

Holding period is the number of months between the entry and exit of the LBO sponsor from the LBO. It is based on a 30 day month. The going private LBO is represented by a dummy variable that assumes a value of 1 if the LBO prior to exit was originally a publicly listed company and 0 otherwise.

The sample

Our initial list of 275 LBO exits is from CMBOR quarterly review publications covering the period from spring 1998 to spring 2004. We obtained the LBO target name, the year, deal value, and exit prices from this source. CMBOR holds a comprehensive database, on management buy-outs and buy-ins in the UK. After excluding 8 companies that were exited through a receivership, and 33 others because of undisclosed deal value or exit value, we tried to obtain further details (identity of the LBO sponsor and exact deal month) on the remaining 234 LBOs. Such information is available from the deal synopsis in Thompson's SDC database, Wall Street Journal and Financial Times (through the Factiva press database), and also the investment track record/case studies on the websites of the private equity firms. A further 96 were excluded because we could not get information on the deal, and therefore could not ascertain the private equity firm involved. Of the 138 firms left, we dropped 29 for which we could not verify the date of the deals. we then went ahead to extract financial data of the 109 firms from FAME database, and dropped another 5 firms for which the financial information for the year required was not found in the FAME database, leaving a final sample of 104 LBOs.

The dates and values in the samples were also verified from the deal synopsis, and the investment profile published on the websites of the private equity firms. Data on IPO volumes and values were downloaded from the London Stock Exchange website, and includes both the Official List flotations and flotations on the Alternative Investment Market (AIM) while data on the corporate M&A deal flow (volumes and values) were obtained from Thompson's SDC database.

In collecting the financial data of the sample companies, we used the latest published financial statement of the company, on a date preceding the exit of the LBO sponsor. In a few cases where we got only the deal month and year as against the exact date, we assumed the deal to have taken effect from the first day of the month. We believe that this would not have any significant effect on the annualised return multiple. Also where there is a change in reporting date resulting in the publication of results for a period other than 12 months, the operating period data (i.e. EBIT, depreciation and amortisation) are re-based to a full year.

Sample description

The sample constitutes of 104 UK LBO target firms. Table 1 shows the distribution of the sample by industry. It could be seen that most of the sample (72%) is concentrated on four industry sections; 27 (26%) of the sample were manufacturing companies, while another 25 (24%) were from the real estate, renting and business activities industry sector. Another 13 (12.5%) were from the wholesale, retail and repairs section and then 10 (9.6%) are from the social and personal services sector. It is also apparent that certain sectors are either not represented or have very few observations in my sample; these include fishing, mining and quarrying, electricity gas and water supply, public administration, defence and social security, education and agriculture. These are either capital intensive or not so profitable industry sectors. This is conceivable for leveraged buyouts because the investors are financial buyers whose ultimate motivation is profit.

Table 1 here

A total of 21 firms (20%) were previously public companies before the buyout, and only 5 (24%) of them were exited by a reverse LBO. Trade sale seems to be most popular, with 12 (57%) exits. Most flotation exits were in the business activities

category that consists mostly of high-tech, computer and research companies. Figure 3 shows the sample distribution by exit route and illustrates the dramatic growth of exits by trade sale, peaking in 2000 dropping gradually from 2001 to 2002, while secondary buyouts continued to grow, resulting in a switch in relative volumes between both routes over the period from 2000 to 2003. This same trend is evidenced in Figure 1 which captures the distribution of the larger population from which the sample was selected. 81 (78%) of the firms were exited after the internet bubble in April 2000. Of the 33 secondary buyouts, 27 (81.2%) were after April 2000. Figure 4 illustrates the IPO activity on the London Stock Exchange over the period under review. It shows the steady decline in volumes from 2000 through 2003 together with the decline in amounts raised from 2002.

4. Results

4.1 Descriptive statistics

Summary statistics of the data is presented in Tables 2 while Table 3 shows the summary statistics of differences between groups. From Table 2, the mean total assets of companies in the whole sample are £183 million, and the median total assets of £41 million show that the sample consists mainly of smaller sized companies. The average holding period is 40.5 months, with secondary buyouts (SBO) having been held longest, for an average of 47.6 months before exit. Table 3 (Panel B) shows that difference in mean holding periods is only statistically different between secondary buyout and trade sale but the median holding period in a SBO is significantly longer than with an IPO or trade sale.

Tables 2 and 3 here

The mean and median EBITDA/Total assets for the whole sample are 0.17 and 0.16 respectively. While there is no significant difference in these central tendency measures of operating performance between trade sale and flotation, the differences are statistically significant at 5% between flotation and SBO and at 10%, between SBO and trade sale (Table 2, Panel D). This suggests that firms in SBOs are relatively better in operating performance (mean and median = 0.24) than those exited through the other

two routes. The mean debt/total assets for the whole sample is 0.48, but Table 2, Panel B shows the average financial leverage level of firms exited through flotation is 0.57, though there is no statistically significant difference between groups as shown in Table 3, Panel E.

The whole sample mean annualised exit return is 79% (median 25%), flotation taking the lead with a mean of 116% (34%) and trade sale giving the least mean return of 43% (21%). But Table 3 Panel F shows that the differences in their means are not statistically significant¹⁰ except between the mean returns of IPOs and trade sales (significant at 10%).

Overall, SBOs are characterised by the best operating performance and the longest holding period among the three exit types. The highest returns are however generated by IPO exits.

4.2 Empirical results of the trinomial logistic regression

Table 4 presents the parameter estimates of the trinomial logistic regression model. The global chi-squared test for the significance of the predictor set is 42.8 with 20 degrees of freedom which is highly significant (p -value = .002) (Liao, 1994; Demaris, 1992 and Aldrich and Nelson, 1984). The logistic regression results show the overall prediction accuracy of the model to be 55.8%. In evaluating the model's accuracy in classifying the sample into the three exit route categories, we employ the evaluation criteria suggested by Anderson and Black (1998). The approach is to determine the percentage of the sample that could have been classified correctly by chance and then compare this to the classification accuracy achieved by the model. The proportional chance criteria is used when the group sizes are unequal (as in this case; flotation= 32, trade sale= 39 and secondary buyout=33), and the interest is to correctly identify members of each group. It is given by the formula:

$$C_{pro} = (p_{flotation})^2 + (p_{trade\ sale})^2 + (p_{secondary\ buyout})^2$$

¹⁰ To cross validate this finding, we applied the 'Eta squared' effect size statistics. Effect size statistics provides an indication of the magnitude of the differences between the groups (not just whether the difference could have occurred by chance) (Pallant, 2001). Eta squared = $t^2 / \{t^2 + (N_1 + N_2 - 2)\}$ where t is the t -value from independent t -test, and N_1 and N_2 represent the individual sample size of both groups. The Eta statistics showed that there is only small effect size between the paired comparisons of the means

Where $p_{\text{flotation}}$, $p_{\text{trade sale}}$ and $p_{\text{secondary buyout}}$ represent the proportion of the total sample belonging to the respective exit route categories. Therefore, C_{pro} for the model is

$$C_{\text{pro}} = (0.30767)^2 + (0.375)^2 + (0.3173)^2 \\ = 33.6\%.$$

Therefore, going by chance, only 33.6% of the sample would be classified accurately, but the estimated model has 55.8% classification accuracy.

Table 4 here

The second column in Table 4 shows the global tests (χ^2) for the effects of each predictor on the dependent variable (exit route). It indicates the contribution of each variable to the overall relationship between the exit routes and the individual independent variables. The results show that operating performance (EBITDA/Total assets), size of the firm (log of total assets), holding period, high-tech industry effects and leverage play a prominent role in explaining LBO exit route. The logistic coefficients indicate the role of each independent variable in differentiating between the specified exit routes. We only discuss variables that are significant. As regards the choice between flotation and SBO, high EBITDA/Total asset ratio, high total asset level and longer holding period increase the relative probability of exit through secondary buyout whereas high Total debt / Total asset and belonging to a high-tech industry increase the relative probability of an IPO exit.

As regards the choice between a trade sale and an SBO, once again high EBITDA/Total assets, high level of total assets and a long holding period make an SBO exit more likely. Neither leverage nor high-tech industry affiliation affects this choice. Thus the choice between these pairs of exit routes is influenced by overlapping but not identical sets of underlying factors. While strong operating performance, large firm size and long holding period favour an SBO over either an IPO or trade sale, high leverage and high-tech industry affiliation favour the IPO route over the SBO. These additional factors suggest that LBO sponsors may be opportunistically selecting the stock market exit when investor sentiment favours speculative, high-tech investments. These results are consistent with the findings reported by Schwienbacher (2004) and also with the market

of the three exit route returns, and this is further corroborated by the result of the ANOVA discussed in detail below.

timing for appropriate valuation theory of Lerner (1994). The model does not show any significant influence by other variables in differentiating between the exit routes.

4. 2 Empirical results of the One – way ANOVA

A one-way, between-groups analysis of variance was conducted to explore the impact of exit route on the resulting returns, as measured by the annualised exit multiple. Exit return was the dependent variable, and the three exit route categories, the independent. Levene's test for homogeneity of variances returned a significance of 0.152. This is greater than 0.05, therefore, the ANOVA assumption of homogeneity of variances is not violated (Pallant, 2001). The overall ANOVA suggests that there is no statistically significant difference in the exit returns among the three exit routes ($F=0.836$ and $p=0.436$). This is in line with the independent t-test statistics presented earlier (see Table 3). Thus LBO sponsor's choice is dictated by considerations about deal feasibility, financing mix after exit, opportunistic exploitation of investor sentiment about high-tech industries and deal size rather than by the exit return. The flotation route is preferred by sponsors whose firms are relatively small and burdened with high leverage. Plausibly, they use flotation as a means of lowering the leverage.

Robustness tests

To test the robustness of the trinomial logistic model, we re-estimate the model using the pound sterling values of IPOs, and the value of transactions at the corporate M&A market as proxies for exit market liquidity. The results are presented in Tables 5 and 6. The result remained more or less the same (in terms of coefficient signs and significance) for all variables except for log of total assets, which is no longer significant in the flotation vs secondary buyout equation when the value of transactions in the corporate M&A market is used as proxy for exit market liquidity. Both models were highly significant, at $p=0.003$ and maintained the classification accuracy of 55.8%.

We also re-estimated the model without the exit returns to see the impact of its exclusion on the model. The results set out in Table 7 show no significant change in the

parameter estimates. The overall model still remains significant at $p=0.004$, but the model classification accuracy drops from 55.8% to 51%, which is still respectable¹¹.

4.3 Discussion of results

Effect of operating performance and exit returns on exit route

The results of the independent t- tests and ANOVA presented earlier show that there is no statistically significant difference in the exit returns from the three exit routes and between pairs using UK sample. The logistic regression model also suggests that the exit return is not a significant variable in predicting exit route. This puts to question the validity of the pecking order theory in which flotation is preferred because of the relative returns, associated with it as compared a trade sale and secondary buyout. My finding is contrary in principle to that of Bienz (2004), in which statistically significant difference in the mean IRR between flotation and trade sale was found, using German data. This difference may be due to sampling differences. His sample covers a much longer time horizon (1971 - 2003) which would have included longer periods of capital market boom, during which stocks are likely to have been mis-priced (over-valued) as suggested by Ritter (1984, 1991), in which case returns from flotation may be clearly higher than through other routes. On the other hand, however, about four years (April 2000 – June 2004) of the period covered by this study were post-internet bubble with dismal stock market activity. Furthermore, this period is likely to be associated with more severely under-priced flotations to lure investors and this could have put further downward pressure on returns from flotation and perhaps trade sales to a lesser extent.

The model estimates (in Table 4) on the influence of operating performance on LBO exit route is quite interesting and may also throw more light on the finding regarding exit returns. It predicts that bigger firms with better operating performance are more likely to be exited through secondary buyouts rather than flotation or trade sale.

¹¹ On using the EBITDA margin as proxy for operating performance in place of EBITDA to total asset used in the model, the result however differed remarkably. the parameter estimates for log of total assets and EBITDA margin are not significant in the resulting model; instead, the estimates for consolidation prone industries and public to private dummy show marginal significance at 10% level. It is unclear why

The relatively better operating performance of firms exited by secondary buyouts and their longer holding periods, suggests that LBO sponsors hold-on to the better performing firms longer, in anticipation of favourable stock market conditions, to maximise returns. However, when the pressure to meet their financial obligations heightens, they exit by selling to other financial buyers in a secondary buyout. This probably explains the significantly longer holding period, and better operating performance for firms exited by secondary buyout than through flotation and trade sale, and leads me to ask the question; have the best companies been necessarily floated post the internet bubble? In checking whether the longer holding period reflects the time required for them to achieve improved operating performance as noted by Sudarsanam (2003), we found no significant correlation between the holding period and operating performance, instead I found a weak inverse relationship ($r = -0.07$). The correlation is probably weakened by the relatively underperforming companies with equally long holding periods that characterise the flotation sample.

Furthermore, the usually shrewd financial buyers operating in the secondary buyout market would have the skills and experience to do a thorough due diligence to avoid buying a 'lemon' (Akerlof, 1970). Therefore, they seek out high performance companies and provide needed exit liquidity to the existing LBO sponsors. The result of this could be a systematic sieving-off of the best firms from the portfolio companies of LBO sponsors during periods of relative low market liquidity, resulting in the lower operating performance firms being packaged and possibly "window dressed" for flotation, in line with the performance manipulation hypothesis advanced by Degeorge and Zeckhauser (1993). Through this mechanism, LBO sponsors reserve the best companies for disposal in the secondary market, as only these companies are likely to scale through the more rigorous due diligence and health checks of the secondary buyer.

I also present another possible explanation for the higher operating performance, from an ownership and corporate-control point of view. The owner-managers of companies, with high current operating performance and potential for similar future performance, may desire to retain control and undiluted equity structure, and therefore support/encourage LBO sponsor to exit through secondary buyout.

this would be the case, but a possible explanation would be that EBITDA margin may show some industry biases, and my sample consists of firms from a wide range of industries.

Also worthy of note is the argument that secondary deals keep money going round in the private equity industry, especially when other exits like IPOs are weak. “The growing trend toward secondary deals also raises questions over the private-equity industry’s fee structure. Sponsors earn most of their profit when assets are sold and funds returned to investors. There is a potential conflict of interest in them shuffling assets between themselves” (Dixon, 2004)¹². Analysis of these aspects may provide more insights into other drivers of the growth of the secondary market.

An important factor that may be driving the growth of the secondary market in LBOs and possibly its profile in terms of exit returns that are comparable to other exit routes may be the fact that many investors are now viewing private equity as an asset class that should be actively managed and therefore dynamically re-balancing their private and alternative investments relative to other asset classes to comply with investment mandates and to reflect their dynamically changing market outlook (Clark and Kojima, 2003). These interesting dimensions may be an indication that secondary buyouts may no longer be as relegated as it has been in the pecking order.

The effect of financial leverage and size

The estimated model (in Table 4) predicts that the higher the financial leverage of UK LBO’s, the more likely they are to be exited through flotation and trade sale (though the estimates for trade sale is not statistically significant) than secondary buyout. This finds support in the arguments put forward by Pagano et al (1998), that highly leveraged firms may consider an IPO as a means of reducing their debt burden. It is also consistent with the findings of Brau et al (2003) that higher average debt ratios increase the likelihood of exit through flotation as against a trade sale.

However, contrary to expectations, and the findings of Pagano et al (1998), the model predicts that the larger the firm size, the less likely it is to be exited through an IPO and a trade sale. By inference, smaller firms are more likely to go public, and this is conceivable, as highly geared small firms with high growth potentials and large future capital investment plan would be left with little choice but to finance such investments with funds from the stock market. The introduction of the junior market for small firms, AIM, as part of the London Stock Exchange in 1998 has provided the platform for even

¹² Trend Towards Secondary Deals Grows, by Hugo Dixon, Wall Street Journal Europe; Monday July 5, 2004.

the very small firms to actualise their dreams. The AIM now provides a platform for cheaper and less strenuous flotation, with relatively minimal disclosure and reporting requirements for small firms.

Another possible explanation for the lower propensity for flotation by larger firms is that, while the capital market was generally bearish from 2000 to early 2004, it was more favourable to smaller capitalisation firms in terms of fund raising, as is evidenced by the continued drop in the average deal size. Desperate for funds to finance growth and expansion, smaller companies could easily settle for relatively higher IPO discounts that investors are likely to have demanded.

In accordance with expectations, the model predicts that firms in the high-tech industries are more likely to be exited through flotation than secondary buyout. But the dummies for consolidated industries, LBO sponsor reputation and public to private LBOs were not significant. The parameter estimates for market liquidity proxies in all models were not statistically significant. This is probably because the bulk of the data (78%) were in a period of predominantly low market liquidity, thereby undermining the predictive power of this variable on the exit routes.

5. Summary and Conclusions

A critical challenge facing any private equity firm in an LBO is that of achieving a successful exit at an appropriate time so as to achieve maximum returns. Ensuring this requires a proactively articulated exit strategy. Whereas previous studies mostly using data from the U.S. and other European countries have documented the importance of some factors in determining the ultimate exit route, there is no known UK study that addresses the predictive variables critical to this strategic decision. Furthermore, the huge size of the UK LBO market, which is the largest in Europe (at over £16 billion in 2004), and the influx of foreign private equity operators in recent years makes this study inherently interesting.

The study provides evidence on determinants of exit route using 104 going concern UK LBOs exited between January 1998 and June 2004, by estimating trinomial logistic regressions with the three common exit routes for such businesses (flotation through IPO, trade sale and secondary buyout) as the response variables. The results show that operating performance, firm size, length of LBO investment holding period

and whether the firm belongs to the 'high-tech' industry are significant determinants of the exit strategy. Operating performance is significantly higher in secondary buyout sample than in the other two categories and holding period is also significantly higher in secondary buyout than in trade sale. Though the evidence might not be conclusive, it suggests that when exit markets are not favourable, LBO sponsors hold on to better performing firms longer, in anticipation for favourable exit market conditions, while disposing of the others through trade sale and flotation. This is the first study that indicates that the secondary LBO market might be 'sieving-off' the best companies from the portfolios of LBO sponsors leaving the others to be sold to stock market investors or strategic buyers.

This study also makes a contribution to existing theory on the effect of firm size and exit returns on exit route decision. It finds that exit returns have no significant predictive power on the exit route, and that larger firms less likely a firm is to be exited by flotation and trade sale. These are at variance with existing theory which holds that larger firms are more likely to be exited by flotation, and that flotation gives higher returns than other routes. We interpret my finding to be the result of the unfavourable market conditions which was prevailing during the period when over 78% of the sample firms were exited (post-internet bubble). During times of unfavourable market condition, smaller firms tend to be more active in fund raising from the stock market than larger firms.

Limitations and further research

1. A longer time horizon would have made for a more reliable estimate of the predictive power of exit market liquidity on the exit strategy, as against the period we have used which was mostly a down-time for the market.
2. Due to dearth of data, exit multiples used have ignored the possibility of intermediate cash flows. The LBO firm may have received significant dividends during the holding period which may not have been captured, and therefore in some cases where I have computed a negative return, it may not actually be negative had I included the effect of the dividends. Also the sponsor may have made post-LBO investments e.g. capital expenditures. Our measure ignores these and therefore may overstate the exit returns. However, there is no *a priori* reason for these intermediate

cash flows to differ systematically across the exit categories. Thus our exit returns may still be unbiased measures of true returns.

3. The use of ex-post exit returns as proxy for ex-ante returns is not ideal, even though it did not have any noticeable effect on the model estimates. Once again this is unlikely to bias the results in favour of a particular exit route.

The evidence presented in this study provides significant insights and opens the grounds for potentially fruitful research in the future. Future studies may want to repeat this study over a longer time horizon, to capture a wider spectrum of market liquidity cycle, and of course a larger sample size. Also, another area of potential research would be to capture the value added by LBOs through improved corporate governance structures etc, and consider the relationship of those to exit returns.

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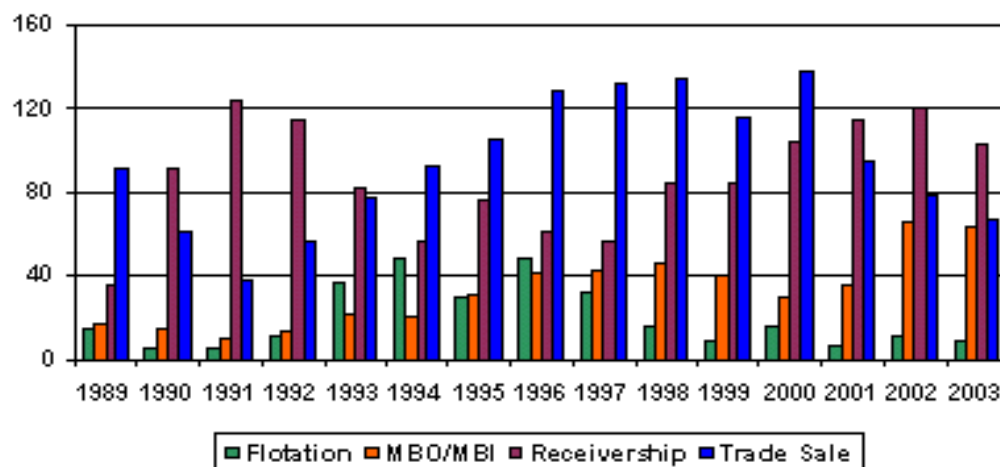
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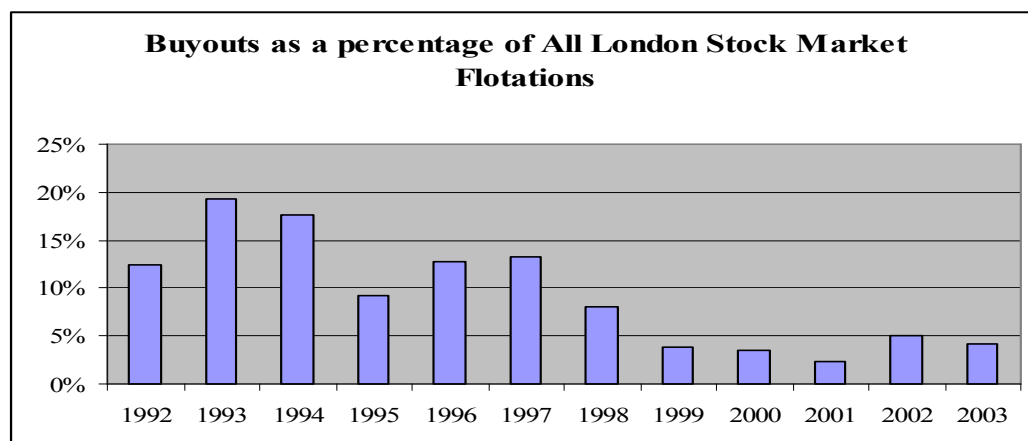
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Figure 1: Exits of UK Buy-outs and buy-ins from 1989 – 2003.



Source: CMBOR Quarterly Review, Spring 2004.

Figure 2: Buy-outs as a percentage of London Stock Market Flotations, 1992-2003



Source: CMBOR Quarterly Review, Spring 2004.

Figure 3: Sample distribution by exit route over the period (Jan. 1998 – June 2004).

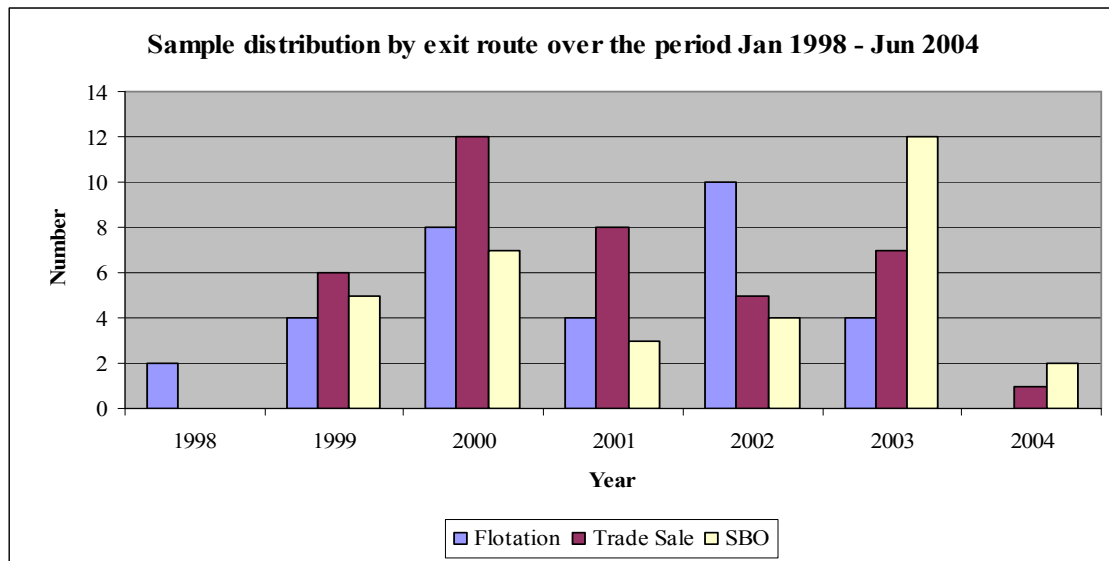
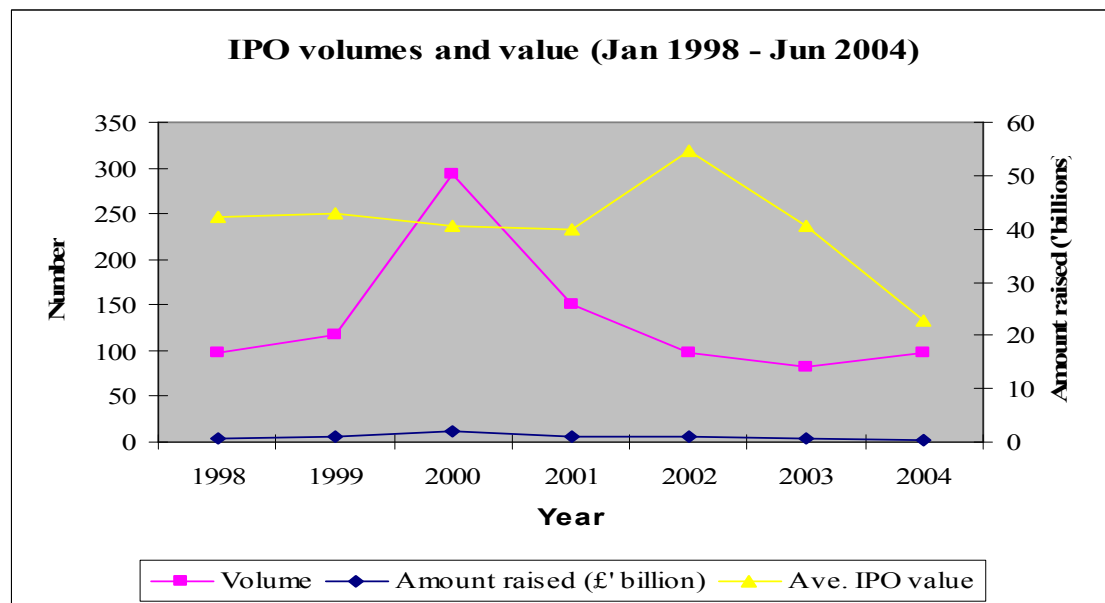


Figure 4: London Stock Exchange IPO volumes and values (1998 - 2004)



Source: Thomson SDC Platinum database.

Table 1: Sample distribution by industry

Table I shows the distribution of the whole sample by industry, exit route and whether the company was public or private prior to the buyout. The sample has been sorted by industry on the basis of the two-digit UK SIC 2003 codes. The third column shows the total sample in each industry section, while the last six columns show the distribution by exit route, split into whether previously public or privately held companies before the buyout.

Industry	Total	Flotation		Trade sale		Secondary buyout	
		Public	Private	Public	Private	Public	Private
A) Agriculture, hunting and forestry	1	0	0	0	0	0	1
D) Manufacturing	27	-	4	2	7	2	12
F) Construction	3	0	0	2	0	0	1
G) Wholesale and Retail Trade; Repair of Motor Vehicles, Motorcycles and Personal Household Goods	13	3	1	2	4	0	3
H) Hotels and Restaurants	6	0	3	0	1	1	1
I) Transport, Storage and Communication	6	0	0	0	4	0	2
J) -Financial Intermediation	5	0	3	1	1	0	0
K) Real Estate, Renting and Business Activities	25	1	13	4	4	0	3
M) Education	1	0	0	0	1	0	0
N) Health and Social Work	7	0	2	1	2	0	2
O) Other Community, Social and Personal Service Activities	10	1	1	0	3	1	4
Total	104	5	27	12	27	4	29

Table 2: Descriptive Statistics

The summary statistics in Panel A refers to the entire sample, in Panel B to the sample companies that were exited through flotation, in Panel C to sample companies that were exited through a trade sale, and in Panel D sample companies that were exited through a secondary buyout.

Variable	Mean	Median	Sample size
Panel A: The whole sample			
Total asset (£'million)	182.70	40.67	104
Holding period (months)	40.59	37.00	104
EBITDA /Sales	0.17	0.16	104
EBITDA/Total asset	0.18	0.18	104
Total debt/Total asset	0.48	0.41	104
Annualised exit return	0.79	0.25	104
Panel B: Flotation sample			
Total asset (£'million)	218.53	40.91	32
Holding period (months)	38.28	31.50	32
EBITDA /Sales	0.16	0.15	32
EBITDA/Total asset	0.11	0.16	32
Total debt/Total asset	0.57	0.45	32
Annualised exit return	1.16	0.34	32
Panel C: Trade sale sample			
Total asset (£'million)	160.72	38.45	39
Holding period (months)	36.54	36.00	39
EBITDA /Sales	0.17	0.14	39
EBITDA/Total asset	0.17	0.17	39
Total debt/Total asset	0.44	0.40	39
Annualised exit return	0.43	0.21	39
Panel D: Secondary buyout sample			
Total asset (£'million)	173.93	47.08	33
Holding period (months)	47.61	42.00	33
EBITDA /Sales	0.22	0.19	33
EBITDA/Total asset	0.24	0.24	33
Total debt/Total asset	0.44	0.35	33
Annualised exit return	0.85	0.25	33

Table 3: Difference tests for variables between exit routes

Independent-sample t-test is used to investigate the difference in means between pairs of exit route samples for relevant variables, while the Mann-Whitney U Test is used to compare the medians for the same variables. The statistics are set out in Panels A – F for total asset, holding period, EBITDA margin, EBITDA/total asset, total debt/total assets and annualised exit returns respectively. Where the significance level of the Levene's test is greater than 0.05, I report the "Equal variances assumed" t - values and where it is ≤ 0.05 , I report the "Equal variance not assumed" t - values, and these are denoted with * indicating significance of Levene's test at 5%.

Variable	Mean		Median	
	T - value	p-value	z - value	p-value
Panel A: Total asset				
Flotation and Trade Sale	0.53	0.60	-0.06	0.95
Flotation and Secondary buyout	0.48	0.64	-0.97	0.33
Trade Sale and Secondary buyout	-0.15	0.88	-1.32	0.19
Panel B: Holding period				
Flotation and Trade Sale	0.32	0.75	-0.23	0.82
Flotation and Secondary buyout	-1.40	0.17	-1.91	0.06
Trade Sale and Secondary buyout	-2.01	0.05	-1.89	0.06
Panel C: EBITDA/Sales				
Flotation and Trade Sale	-0.67	0.51	-0.22	0.83
Flotation and Secondary buyout	-1.00	0.32	-0.85	0.40
Trade Sale and Secondary buyout	-1.13	0.26	-1.11	0.27
Panel D: EBITDA/Total Assets				
Flotation and Trade Sale	-1.07	0.29	-0.33	0.74
Flotation and Secondary buyout	-2.05	0.05	-2.34	0.02
Trade Sale and Secondary buyout	-1.97	0.05	-1.92	0.06
Panel E: Total debt/Total asset				
Flotation and Trade Sale	1.08	0.29	-0.67	0.51
Flotation and Secondary buyout	1.19	0.24	-0.62	0.54
Trade Sale and Secondary buyout	0.06	0.96	-0.17	0.87
Panel F: Annualised exit return				
Flotation and Trade Sale	1.66	0.10	-1.10	0.27
Flotation and Secondary buyout	0.42	0.67	-1.09	0.28
Trade Sale and Secondary buyout	-0.75	0.46	-0.12	0.91

Table 4: Trinomial logistic regression

The table presents the logistic regression models. The dependent variables are the three exit routes – Flotation through IPO, Trade sale and Secondary buyout (SBO). SBO is the reference category. * Indicates significance at 5% and ** at 10%.

Variable	χ^2	Flotation vs SBO		Trade Sale vs SBO	
		B	Standard Error	B	Standard Error
Intercept	9.98*	7.18*	2.75	6.72	2.73
EBITDA/Total asset (OPPERF)	6.66*	-5.60*	2.44	-5.35*	2.35
Annualised exit return (EXITRTN)	4.59	-0.19	0.15	-0.34	0.26
Log of total assets (FSIZE)	5.01*	-0.44**	0.23	-0.43**	0.22
Total debt/Total asset (FINLEVER)	3.27	1.50**	0.84	0.74	0.78
LBO Sponsor reputation (PEREPUT)	2.90	-0.02	0.25	0.32	0.24
Log of holding period (HPERIOD)	8.81*	-1.43*	0.60	-1.44*	0.58
IPO volume (MKTLIQ)	2.04	-0.00	0.00	0.00	0.00
High-tech dummy (FINDUST)	6.95*	1.86*	0.79	0.50	0.81
Consolidation industries (FINDUST)	2.87	0.96	0.73	-0.17	0.74
Public to private LBO dummy (PTPS)	3.02	0.60	0.86	1.24	0.76

Model fit and accuracy

X ² for model	42.83	Pseudo R-squared (McFadden)	0.19
p-value for model	0.002	Model overall classification accuracy	55.8%

Table 5: Trinomial logistic regression with IPO value of activity

The table presents the logistic regression models. The dependent variables are the three exit routes – Flotation through IPO, Trade sale and Secondary buyout (SBO). SBO is the reference category. * indicates significance at 5% and ** at 10%. In this model version, the proxy for market liquidity is the log of IPO value as opposed to the IPO volumes used in Table 4.

Variable	χ^2	IPO vs SBO		Trade sale vs SBO	
		B	Standard Error	B	Standard Error
Intercept	5.821*	6.107*	2.958	5.867*	2.908
EBITDA/Total asset (OPPERF)	6.051	-5.310*	2.423	-5.293*	2.370
Annualised exit return (EXITRTN)	4.480	-0.061	.153	-0.363	0.274
Log of total assets (FSIZE)	4.423	-0.397**	.230	-0.417**	0.223
Total debt/Total asset (FINLEVER)	3.210	1.451**	0.836	0.697	0.803
LBO Sponsor reputation (PEREPUT)	2.968	-0.055	0.245	0.298	0.234
Log of holding period (HPERIOD)	8.101*	-1.372*	0.596	-1.409*	0.584
Log IPO value (MKTLIQ)	0.773	0.126	0.730	0.530	0.659
High-tech dummy (FINDUST)	6.420*	1.810*	0.794	0.568	0.806
Consolidation industries (FINDUST)	3.195	1.052	0.735	-0.141	0.745
Public to private LBO dummy (PTPS)	3.097	0.551	0.869	1.236	0.763

Model fit and accuracy

χ^2 for model	41.573	Pseudo R-squared (McFadden)	.183
p-value for model	.003	Model overall classification accuracy	55.8%

Table 6: Trinomial logistic regression using log of corporate M & A deal values

The table presents the logistic regression models. The dependent variables are the three exit routes – Flotation through IPO, Trade sale and Secondary buyout (SBO). SBO is the reference category. * Indicates significance at 5% and ** at 10%. In this model version, the proxy for market liquidity is the log of the M&A deal value as opposed to the IPO volumes used in Table 4.

		IPO vs SBO		Trade sale vs SBO	
Variable	χ^2	B	Standard Error	B	Standard Error
Intercept	1.543	5.027	4.299	4.011	4.010
EBITDA/Total asset (OPPERF)	6.261*	-5.351*	2.418	-5.398*	2.372
Annualised exit return (EXITRTN)	4.515	-0.068	0.154	-0.366	0.276
Log of total assets (FSIZE)	3.975	-0.383	0.234	-0.401**	0.225
Total debt/Total asset (FINLEVER)	6.261	1.455**	0.836	0.696	0.806
LBO Sponsor reputation (PEREPUT)	3.048	-0.058	0.244	0.299	0.233
Log of holding period (HPERIOD)	7.726*	-1.357*	0.598	-1.388*	0.587
Log of M&A value (MKTLIQ)	.923	0.261	0.680	0.574	0.608
High-tech dummy (FINDUST)	6.396*	1.812*	0.796	0.579	0.808
Consolidation industries (FINDUST)	3.352	1.064	0.732	-0.159	0.742
Public to private LBO dummy (PTPS)	3.050	0.542	0.873	1.226	0.764

Model fit and accuracy

χ^2 for model	41.722	Pseudo R-squared (McFadden)	.183
<i>p</i> -value for model	.003	Model overall classification accuracy	55.8%

Table 7: Trinomial logistic regression without exit returns

The table presents the logistic regression models. The dependent variables are the three exit routes – Flotation through IPO, Trade sale and Secondary buyout (SBO). SBO is the reference category. * Indicates significance at 5% and ** at 10%. In this model version we exclude the exit return included in Table 4.

Variable	χ^2	IPO vs SBO		Trade sale vs SBO	
		B	Standard Error	B	Standard Error
Intercept	9.658*	7.503*	2.783	5.9878*	2.613
EBITDA/Total asset (OPPERF)	6.650*	-5.688*	2.419	-5.175*	2.328
Log of total assets (FSIZE)	6,272*	-0.485*	0.234	-0.472*	0.217
Total debt/Total asset (FINLEVER)	3.767	1.579**	0.840	0.893	0.771
LBO Sponsor reputation (PEREPUT)	2.711	-0.034	0.246	0.291	0.231
Log of holding period (HPERIOD)	8.091*	-1.468*	0.588	-1.175*	0.547
Log IPO volume (MKTLIQ)	1.368	-0.004	0.004	0.000	0.003
High-tech dummy (FINDUST)	7.444*	1.864*	0.792	0.416	0.786
Consolidation industries (FINDUST)	3.504	1.046	0.740	-0.187	0.729
Public to private LBO dummy (PTPS)	2.378	0.477	0.826	1.034	0.713

Model fit and accuracy

χ^2 for model	38.24	Pseudo R-squared (McFadden)	0.17
<i>p</i> -value for model	.004	Model overall classification accuracy	51%